

Extraction of PDF Table Data Based on the Pdfplumber Method

Wen Yang*

Shandong Yingcai University;
Shandong Key Laboratory of Electric
Power Big Data Processing and
Security
China
yw501@126.com

Feifei Cao

Shandong Yingcai University;
Shandong Key Laboratory of Electric
Power Big Data Processing and
Security
China
caofeifei_2011@163.com

Xueli Zhao

Shandong Yingcai University;
Shandong Key Laboratory of Electric
Power Big Data Processing and
Security
China
17863990157@163.com

Abstract

A large amount of information is stored in various electronic documents, including scientific literature, reports, and research papers. The table in these documents is one of the main ways to contain rich and critical information. To edit and reuse the table data in PDF files, automatic recognition and extraction of PDF table data is implemented based on the pdfplumber parsing method. First, the pdfplumber converts PDF files into the parsable Python objects, including text, images, tables, and other information in the file. Then it parses the specified PDF pages in the Python object, converts them into the table data, and automatically extracts them into Excel files. The experimental results show that the integrated process can identify and extract the table data in PDF files in batches. Using the recognition method of pdfplumber, the recognition rate of table data such as the survey tables and the education resource tables in higher education quality reports reaches an average of 96%, effectively handling complex situations such as blank spaces and column misalignment in the tables.

CCS Concepts

• **Applied computing**; • **Document management and text processing**;

Keywords

PDF files, pdfplumber, Table data

ACM Reference Format:

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1 Introduction

With the continuous development of society, market competition is becoming increasingly fierce, and the demand for efficient office

work in all walks of life is becoming increasingly urgent [1]. Before the birth of electronic documents, documents generally existed in the form of printed paper, which was very inconvenient for moving, storing, retrieving, and backing up.

After the emergence of electronic documents, paper documents are scanned into images and stored in them, and tables naturally exist in the form of images in large quantities. PDF files have become a widely used electronic document format [2]. Among them, PDF tables are a common form of data in PDF files, which often contain a large amount of information such as data, text, charts, etc [3].

The demand for tables is not only for transmission and storage, but also for reuse. How to recognize the tables in electronic documents and extract their contents has become a key to the reuse of table contents [4]. The tables in PDF files may have different layouts and structures, which makes it difficult to accurately identify the boundaries and cells of tables from the text. The extraction and processing of these table data has become an important research direction in the field of data processing [5].

This article aims to explore the relevant technologies and methods for extracting PDF table data based on the pdfplumber method, with a view to providing support for subsequent data processing, analysis and decision-making.

2 The principle of pdfplumber

The pdfplumber is a PDF document parsing method based on the pdfminer and completely developed in Python [6]. It provides a convenient way to process PDF files. The method can not only obtain specific information about each character, rectangular box, line, and other objects, but also be used to extract, analyze, and modify text and table data in PDF files. Its basic principle mainly involves two main aspects:

2.1 Converting PDF files into parsable Python objects

When the pdfplumber loads a PDF file, it converts the file into a parsable Python object. This object contains all the information in the file, such as text, images, tables, and so on. The PDF file itself contains one or more byte streams (i.e., binary data blocks) that contain all the information for the PDF pages, and The pdfplumber converts these byte streams into Python objects.

2.2 Analyzing the content of PDF pages

The pdfplumber converts each page of a PDF document into text or table data by parsing the Python objects. In order to analyze the

*Corresponding author.

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text content on a PDF page, the pdfplumber first converts each page of the PDF document into a bitmap image and uses the technology of computer vision to detect and recognize the text. After that, the pdfplumber applies some natural language processing techniques to extract and organize the text information.

2.3 Advantages

Currently, the pdfplumber only supports editable PDF documents. Although the pdminer can also parse editable PDF documents [7], the pdfplumber has the following advantages:

- (1) Easy to use: the pdfplumber provides a more concise and straightforward API interface, making it easier for users to parse PDF documents, while the API interface of the pdminer is relatively complex and requires more learning and configuration.
- (2) Better table extraction results: the pdfplumber not only achieves good extraction results for full-line tables, but also outperforms the pdminer for less-line and no-line tables.
- (3) Object information is easier to obtain: the pdfplumber can obtain specific information about each character, rectangular box, line, and other objects, which may be difficult to obtain in the pdminer.
- (4) More user-friendly: The packaging and processing of the pdfplumber makes the resulting objects easier to use and more user-friendly. In contrast, the pdminer may require users to have more expertise to fully utilize its features.

3 Recognition and extraction steps of PDF table data

3.1 Batch obtaining file paths

Define a function to batch retrieve PDF files, the function parameter is the folder address, the return value list contains the file names and paths of all PDF files, and reads each PDF file in turn through a for loop to perform table data recognition and extraction. The core code is as follows:

```
def pdfFilePath(path):#A function for obtaining PDF files in bulk
    filePaths = []
    for root, dirs, files in os.walk(path):
        for file in files:
            filePaths.append(os.path.join(root, file))
    return filePaths
for path in pdfFilePath(filepath):
    extract_table(path)#A function for recognizing and extracting
    table data
```

3.2 Open the PDF file

First, you need to use the pdfplumber method to open the PDF file. By calling the open() method of the pdfplumber, you can obtain a handle to the PDF file, as shown below:

3.3 Locate the table area

Before extracting PDF table data, you need to use the pages[i] method to locate the i-th page where the table is located, and return the information of the current page [8]. The pages are a list

containing page information. Each pdfplumber.page class contains several main attributes:

3.4 Recognize and extract table data

Using the extract_tables() method of the pdfplumber, you can directly extract all tables on the current page. The method returns a list containing all table data, where each table data is a dictionary-type data structure.

By default, the extract_tables uses the vertical and horizontal lines (or rectangular edges) of the page as cell separators, but this method can be customized through the table_settings parameter. Possible settings and their default values are:

```
{
    "vertical_strategy": "lines",
    "horizontal_strategy": "lines",
    "explicit_vertical_lines": [],
    "explicit_horizontal_lines": [],
    "snap_tolerance": 3,
}
```

For extracting regular tables, generally set both vertical_strategy and horizontal_strategy to "lines", and any parallel lines within a snap_tolerance of 3 pixels will be "snapped" to the same horizontal or vertical position. The main code is as follows:

```
table_settings = {"snap_x_tolerance": 3}
tables = page.extract_tables(table_settings)
```

3.5 Save the extraction results

Since the original tables may contain invalid characters such as tabulators, line breaks, and spaces, the preliminary recognition results need to be processed using the replace() method. The processed table data is then stored using the to_excel() method [9], as shown below:

```
table.replace('t', '\t', regex=True).replace('n', '\n',
regex=True).replace(' ', '\s', regex=True)
table.to_excel(excel_writer, index=False, sheet_name=f'Page{j + 1}
table{i + 1}')
```

4 Experiment and Result Analysis

To verify the effectiveness of the above method for recognizing PDF table data, 10 PDF files were selected from the Higher Education Quality Report, each containing six tables to be extracted, and these tables were tested. As shown in Figure 1, one of the questionnaires was about satisfaction.

According to the extraction steps described earlier, the results of table data extraction are shown in Figure 2.

As can be seen from the above figure, most of the table text and data have been recognized and extracted, but there is a column of empty values between columns, and there is also a row of data with column misalignment. There are two ways to solve this column misalignment problem.

4.1 Filling the empty values

Fill the empty values in the 2022 and the investigation number columns in the table with the values in the previous column, as shown in the following code:

Table 1: Pdfflumber.page class properties

Attribute	Description
.page_number	The nume of the page
.width	The width of the page
.height	The height of the page
.objects / .chars / .lines / .rects	Each of these attributes is a list, and each list contains a dictionary that is embedded in each such object on the page

Index	Unit	2022	Investigation Number
Student satisfaction	%	96.96	2000
Among them: Satisfaction with classroom education	%	96.96	2000
Satisfaction with extracurricular education	%	96.69	2000
Satisfaction with the teaching of political courses	%	96.96	2000
Teaching satisfaction of public basic courses (excluding ideological and political courses)	%	95.36	2000
Satisfaction with the teaching of professional courses	%	93.38	2000
Graduate satisfaction	—	—	
Among them: The satisfaction of fresh graduates	%	98.30	3493
Graduate satisfaction within three years after graduation	%	98.00	3493
Staff satisfaction	%	93.42	745
Employer satisfaction	%	99.00	3493
Parent satisfaction	%	98.00	650

Figure 1: Satisfaction survey table.

Index	Unit			2022			Investigation Number
Student satisfaction	%			96.96			2000
Among them: Satisfaction with classroom education	%			96.96			2000
Satisfaction with extracurricular education	%			96.69			2000
Satisfaction with the teaching of political courses	%			96.96			2000
Teaching satisfaction of public basic courses (excluding ideological and political courses)			95.36			2000	
Satisfaction with the teaching of professional courses	%			93.38			2000
Graduate satisfaction	—			—			
Among them: The satisfaction of fresh graduates	%			98.30			3493
Graduate satisfaction within three years after graduation	%			98.00			3493
Staff satisfaction	%			93.42			745
Employer satisfaction	%			99.00			3493
Parent satisfaction	%			98.00			650

Figure 2: The initial result of table data.

```
table.iloc[:, 6][table.iloc[:, 6].isnull()] = table.iloc[:, 5][table.iloc[:, 6].isnull()]
table.iloc[:, 9][table.iloc[:, 9].isnull()] = table.iloc[:, 8][table.iloc[:, 9].isnull()]
```

The filling results are shown in Figure 3. The missing values caused by column misalignment have been filled, and after filling the missing values, the empty value columns and misaligned columns are deleted.

4.2 Adjusting parameters

The "extract_tables()" method has an important parameter "snap_tolerance", which indicates that parallel lines within "snap_tolerance" pixels will be "captured" to the same horizontal or vertical position. Generally, it is set to 3. Considering that the spacing between columns in this table is relatively large, the

tolerance value is increased to 6, and the recognition result is shown in Figure 4.

As can be seen from the above figure, after adjusting the parameter "snap_x_tolerance", there is no empty value column, and there is no column misalignment problem. The average recognition accuracy of table data in all PDF files has reached 96%. It should be noted that the "snap_x_tolerance" parameter value is not fixed, and it needs to be adjusted according to the distribution characteristics of table data in specific applications.

5 Conclusion

This article mainly discusses the relevant technologies and methods for extracting PDF table data based on the pdfflumber. Experimental verification shows that these methods and technologies can

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Satisfaction with the teaching of political courses	%			96.96		2000
Teaching satisfaction of public basic courses(excluding ideological and political)			95.36	95.36	2000	2000
Satisfaction with the teaching of professional courses	%			93.38		2000
Graduate satisfaction	-----			-----		
Among them: The satisfaction of fresh graduates	%			98.30		3493
Graduate satisfaction within three years after graduation	%			98.00		3493
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Figure 3: The filling result of table data.

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Satisfaction with the teaching of professional courses	%	93.38	2000
Graduate satisfaction	-----	-----	
Among them: The satisfaction of fresh graduates	%	98.30	3493
Graduate satisfaction within three years after graduation	%	98.00	3493
Staff satisfaction	%	93.42	745
Employer satisfaction	%	99.00	3493
Parent satisfaction	%	98.00	650

Figure 4: Results of parameter adjustment for table data.

effectively improve the accuracy of PDF table data extraction. However, some shortcomings were also found in the recognition process, such as the limitations in dealing with complex table structures. With the development and application of artificial intelligence and machine learning technologies, more new technologies and methods may be applied to PDF table data extraction in the future to improve its accuracy and efficiency. Therefore, it is expected that more researchers and engineers will conduct in-depth research and exploration in this field in the future, providing more convenient, accurate, and efficient data recognition and extraction services.

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